MIL-W-21081B
1 NOVEMBER 1963
SUPERSEDING
MIL-W-21081A
13 MAY 1959

MILITARY SPECIFICATION

WIRE, STEEL, HIGH CARBON, ROUND, UNCOATED VALVE SPRING QUALITY

This specification has been approved by the Department of Defense, and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 Scope. This specification covers the procurement requirements for two types of uncoated, round (0.060 to 0.250 inch diameter) high carbon steel wire, uniform in quality, manufactured especially for valve springs and other springs requiring high fatigue resistance.

1.2 Classification. The wire shall be of the following types and conditions as specified in the contract or order, (see 6.2). Wire in diameters from 0.060 to 0.092 inch shall be Type II, soft wire only (see 6.1).

Type I — Oil tempered spring wire for the manufacture of springs which require no subsequent hardening.

Type II[1] — Soft wire, furnished in condition A or B, for the manufacture of springs which are to be hardened after cold forming.

Condition A — Spheroidize annealed at finished size.

Condition B — Spheroidize annealed and lightly drawn.

[1] If condition is not specified, condition A or B may be selected at the option of the spring manufacturer.

2. APPLICABLE DOCUMENTS

The following documents, of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein:

STANDARDS

FEDERAL

Federal Standard No. 66 — Steel: Chemical Composition and Hardenability.

Federal Test Method Standard No. 151 — Metals: Test Methods.

MILITARY

MIL-STD-105 — Sampling Procedures and Tables for Inspection of Attributes.

MIL-STD-163 — Preparation of Steel Products for Domestic (Storage) and Overseas Shipment.

(Copies of documents required by contractors in connection with specific procurement functions should be obtained from the procuring activity as directed by the contracting officer.)

FSC 9505

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:39Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Process. The steel shall be made by the open-hearth, electric, or basic oxygen process.

3.2 Chemical composition. The steel shall comply with the requirements shown on Table I subject to the check analysis tolerances of the applicable table of Federal Standard No. 66. An analysis of each melt of steel shall be furnished by the contractor.

TABLE I. *Chemical composition*

Elements	Analysis
	Percent
Carbon	0.60 to 0.75 [1]
Manganese	0.60 to 0.90 [1]
Silicon	0.15 to 0.35
Phosphorous	0.03 maximum
Sulfur	0.03 maximum

[1] For Type I wire, unless otherwise specified, carbon and manganese may be varied outside the stated limits, provided the mechanical properties specified herein are maintained.

3.3 Surface. The wire surface shall be free from imperfections such as seams, pits, die marks, scratches, and other defects tending to impair the fatigue life of the springs (see 4.4.1 and 4.5.4).

3.4 Decarburization. The wire shall show no completely decarburized areas at the surface and partial decarburization shall not exceed a depth of 0.0015 inch (see 4.5.5).

3.5 Dimensions.

3.5.1 *Diameter.* The wire shall be of the size specified in the contract or order (see 6.2), subject to the permissible variations in diameters shown in Table II.

TABLE II. *Permissible variations in diameters*

Diameter	Permissible variations Plus or minus
Inch	*Inch*
0.060 to 0.148	0.0010
0.149 to 0.177	0.0015
0.178 to 0.250	0.0020

3.5.2 *Roundness.* The wire shall not be out of round more than one-half the total permissible variation shown in Table II.

3.6. Wrapping properties. Type I wire shall be free from cracks after wrapping around a mandrel of the size in Table III.

TABLE III. *Mandrel size for wrapping test*

Wire diameter	Diameter of mandrel
Inch	*Inch*
0.093 to 0.162	Diameter of the wire
0.163 to 0.250	Twice the diameter of the wire

3.7 Tensile requirements.

3.7.1 *Type I.* Type I wire shall conform to the tensile requirements shown in Table IV.

TABLE IV. *Tensile requirements, type I*

Diameter	Tensile strength		Reduction of area minimum
	Minimum	Maximum	
Inch	*psi*	*psi*	*Percent*
0.093 to 0.128	230,000	250,000	40
0.129 to 0.162	225,000	245,000	40
0.163 to 0.192	220,000	240,000	40
0.193 to 0.225	215,000	235,000	40
0.226 to 0.250	210,000	230,000	40

3.7.2 *Type II.*

3.7.2.1 *Condition A* — Type II, condition A wire shall be spherodize annealed at finished size and shall have no cold work in the finished condition. The maximum tensile strength shall be 100,000 p.s.i.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:39Z
Check the source to verify that this is the current version before use.

3.7.2.2 *Condition B* — Type II, condition B wire shall be spheroidize annealed and lightly drawn at least one pass through the dies after the last spheroidize annealing treatment. The tensile strength shall be 100,000 p.s.i. minimum to 130,000 p.s.i. maximum.

3.8 Workmanship. The wire shall not be wavy, kinked, nor contain any defects that might impair its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Lot. A lot shall consist of all wire of the same size, type, and condition, manufactured from the same heat, and submitted for inspection at the same time.

4.3 Sampling.

4.3.1 *Chemical test.* Three samples for chemical test shall be taken from wire representing each heat of steel. Milled samples shall be representative of the entire cross section of the wire.

4.3.2 *Surface test.* Samples of the wire, one foot long shall be taken from each end of each coil for the surface test (see 4.5.4).

4.3.3 *Sampling for other tests.* Ten sample coils shall be selected from each lot for decarburization, wrapping, and tensile tests (see 3.4, 3.6, and 3.7). A sample specimen eighteen inches long, shall be taken from one end of each sample coil. If lot is less than ten coils, a specimen shall be taken from each coil in the lot.

4.3.4 *Visual and dimensional.* A sample of coils shall be selected from each lot in accordance with MIL-STD-105, Inspection Level L-8, Minimum sample seven, acceptance number zero.

4.3.5 *Preparation for delivery.* A sample of coils shall be selected from each lot in accordance MIL-STD-105, Inspection Level L-8, Minimum sample seven, acceptance number zero.

4.4 Inspection.

4.4.1 *Visual and dimensional examination.* Sample of wire selected in accordance with 4.3.4 shall be visually examined to determine compliance to the surface requirement of 3.3 (see 4.5.4). The wire shall be visually and dimensionally examined in accordance with 4.4.2 and 4.4.3 to determine conformance to dimensional requirements of 3.5 and the workmanship requirements of 3.8.

4.4.2 *Diameter.* If accessible, the wire shall be gaged near each end and near the middle. If not accessible, a 12 foot length of wire shall be unreeled and gaged in well separated places between the second and twelfth foot from the end. A coil shall be considered noncomforming if the dimensions are not within the specified variations of table II.

4.4.3 *Roundness.* The wire shall be gaged in accordance with the procedures of 4.4.2 and by measuring the diameter twice, 90° apart at the specified locations. A coil shall be considered nonconforming if any two 90° measurements at the same location do not comply with the requirements of 3.5.2.

4.4.4 *Preparation for delivery.* Sample wire coils or coil containers selected in accordance with 4.3.5 shall be visually examined to determine compliance with the

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:39Z
Check the source to verify that this is the current version before use.

preservation, packaging, packing, and marking requirements of this specification.

4.5 Test methods.

4.5.1 *Chemical.* Chemical analysis of samples selected in accordance with 4.3.1 shall be made in accordance with Federal Test Method No. 151, Method No. 111. The sample shall be analyzed for conformity with 3.2.

4.5.2 *Tensile.* Tensile test samples selected in accordance with 4.3.3 shall be tested in accordance with Federal Standard No. 151, Method No. 211. The test specimen shall be straight and the length of the specimen between the jaws of the testing machine shall be not less than 10 inches.

4.5.3 *Wrapping.* Samples selected in accordance with 4.3.3 shall be tested for the wrapping requirements (see 3.6) as follows: Wire small enough to be wrapped by hand may be tested by bending the end of the wire over itself to form a loop with the ends crossing at right angles, the shorter end being several inches long, and then tightly winding the longer end five complete turns about the other, with turns close together. For larger wire, or optionally on any size wire, the wrapping may be done by hand or power device which will wrap the wire closely, around a mandrel of the required diameter (see 3.6) for five complete turns. The sample shall be considered to have failed if any cracks occur in the wire after the first complete turn. The test shall be repeated if a crack occurs in the first turn, since the wire may be bent locally to a radius less than that specified.

4.5.4 *Surface.* The wire shall be etched in a solution of equal parts of hydrochloric acid and water at a temperature of 165° to 175°F. The specimen shall be immersed for a length of time sufficient to remove approximately one percent of the diameter of the wire. The surface of the wire shall be examined after etching under a 10-power binocular microscope for seams, pits, diemarks, scratches and other imperfections

4.5.5 *Decarburization.* Transverse sections of the finished wire samples selected in accordance with 4.3.3 shall be properly mounted, polished, etched, and microstructure examined at a magnification of 100 diameters for conformance to the requirements of 3.4.

4.6 *Rejection and resubmission.* Should any sample fail to comply with the requirements specified herein, the lot shall be rejected. The lot may be resubmitted for inspection provided the contractor has removed defective material or reworked the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Unless otherwise specified in the contract or order (see 6.2) the wire shall be preserved and packaged in accordance with the applicable requirements of MIL–STD–163.

5.1.2 *Level C.* Unless otherwise specified in the contract or order (see 6.2) the wire shall be preserved and packaged in accordance with commercial preservation and packaging methods (see 6.3), adequate to protect the wire from damage and to ensure safe delivery to the destination.

5.2 Packing.

5.2.1 *Level A and B.* Wire preserved and packaged as specified in 5.1.1 shall, where packing is required, be packed in accordance with the applicable requirements of MIL–STD–163.

5.2.2 *Level C.* Wire preserved and packaged as specified in 5.1.2 shall, where packing is required, be packed for domestic shipment in accordance with commercial prac-

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:39Z
Check the source to verify that this is the current version before use.

tice (see 6.3), adequate to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. As a minimum, the shipment shall meet the requirements of carrier rules and regulations applicable to the selected mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order (see 6.2). marking of shipping containers and individual coils shall be in accordance with the requirements of MIL-STD-163.

6. NOTES

6.1 Intended use. Wire covered by this specification represents the best commercially available quality of heat-treated carbon steel wire. This wire is intended mainly for springs subjected to repeated dynamic stresses. Where high quality, Type I, wire is required in sizes under 0.093 inch, music spring wire, Specification QQ-W-470, may be used. Type II wire is generally used only in the larger sizes, however, Type II wire in sizes from 0.060 to 0.092 inch diameter is used in the manufacture of small arms springs.

6.1.1 *Type I.* Type I wire is intended for fabrication into springs which require no subsequent heat treatment except low temperature stress relief.

6.1.2 *Type II.* Type II wire is a soft wire intended for fabrication of springs where forming operations are too severe for Type I wire. Type II wire should be heat treated to a specified hardness after forming.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title and number of this specification.

(b) Diameter of wire required.

(c) Type of wire, and for Type II, the condition (see 1.2).

(d) Quantity of wire required.

(e) Level of packaging and packing (see 5.1 and 5.2).

(f) Special marking if required (see 5.3).

6.3 Commercial practice, preparation for delivery. Commercial practice for preparation of steel products for domestic delivery is defined in Simplified Practice Recommendation 247-52 (Department of Commerce).

Custodians:

Army—MR
Navy—Bureau of Naval Weapons
Air Force—MOA

Other interest:

MI—Review
WE—Review
MU—User
MO—User

Preparing activity:

Navy—Bureau of Naval Weapons

Project No. 9505-0031

Review/user information is current as of the date of this document. For future coordination of changes to this document. draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-19T08:39Z
Check the source to verify that this is the current version before use.